

Pixie Algorithm:

The foundation of pixie-inspired recommendation systems is the notion that significant associations between people and things can be found by employing random walks on bipartite graphs. Users and things (in this case, movies) are the two separate groups of nodes in a bipartite graph, whereas edges show how they interact, such as when a movie is rated. Originally created by Pinterest, the Pixie algorithm explores the local graph structure surrounding a user's preferences to give quick, real-time content recommendations. Pixie makes scalable, effective recommendations by utilizing graph connectedness and local randomization rather than calculating global similarity scores as in conventional collaborative filtering.

The Pixie algorithm's primary mechanism is the random walk. The algorithm moves across the graph by selecting the neighbors at random at each stage, given either a user node or a movie node. It jumps to movies that the user has interacted with, then to other users who have rated those films, and so on, if the walk begins at a user. Therefore, one gets counted for each time a movie node is visited during the walk. Following a walk (of a predetermined length) on this graph, the movies with the highest frequency are deemed to be the most pertinent, and they are then recommended. This method mimics how consumers intuitively look for fresh material based on comparable behavioral patterns and shared interests.

An important benefit of the Pixie-style method is that it's able to uncover mediating relationships, ie relationships that haven't been revealed through direct similarities alone; for example, in the case above, two users who haven't rated a common movie but are connected through a series of common movies and users is a two-hop away, allowing the walk to discover subtle perceptions. This allows the recommender system to offer up movies that a user may never have seen even in the same room of similarity, bringing diversity and novelty to the recommendations. Moreover, the Pixie method scales well to big datasets and allows for real-time personalization as it does not depend on computing or storing global similarity matrices.

An additional benefit is the personalized randomness brought by the walk process. Instead of deterministic algorithms, Pixie can provide slightly different yet relevant recommendations on different runs. Such controlled randomness promotes exploration and prevents echoing chambers that might occur in these overly filtered systems. Through graph connections users are shown both broadly popular and niche content, increasing user satisfaction and engagement. Additionally, there are hyperparameters (such as the length of walks and the number of walks) which are simple to tune by choosing the right amount of exploration or relevance one would like to learn embeddings for.

In real world applications, Pinterest, LinkedIn, and Twitter use graph-based algorithms like Pixie to power recommenders for content, connections, and ads. For instance, Pinterest's Pixie engine produces pin recommendations in milliseconds based on real-time graphs of user-pin relationships. Likewise, LinkedIn employs a connection graph for suggesting jobs or people you might know, and Spotify uses similar strategies to compose playlists in real-time. Through this

project we showed how through the lens of graph based thinking and random-walks we can build a solid foundation for creating lightweight, scalable, and personalised recommendation engines by implementing a Pixie inspired algorithm.